THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME VI

Arbitrary Field Amplitude:
the Langevin Nonlinearity, the Square-Wave Solution,
and the Viscosity Map — Equations (SM-35)–(SM-43)

Where Volume V ended. Volume V completed Sec. II of Shliomis & Morozov. It computed the instantaneous torque (SM-25), separated the fast and slow time scales (SM-26), extracted the rotational viscosity (SM-27), and obtained the central weak-field result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \tag{SM-28}$$

with its zero at $\omega\tau_B = 1$ and its minimum $-\Delta\eta(0)/8$ at $\omega\tau_B = \sqrt{3}$. It then re-derived the same result by the authors' second route and closed the circle by recovering the master formula (SM-34).

Volume VI takes up Sec. III. The restriction $\xi \ll 1$ is abandoned: the Langevin function must now be kept in full, the magnetisation equation becomes genuinely nonlinear, and superposition is no longer available. What survives is (SM-34), because it rests on torque balance alone. The task therefore reduces to computing the averaged torque without approximation — which the authors accomplish by replacing the harmonic field with a square wave, for which the equations can be integrated in closed form. Chapter numbers resume at 50, page numbers at 175.

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Part XI

Arbitrary Field Amplitude

Chapter 50

Equation (SM-35): The Nonlinear Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\left(\boldsymbol{M} - M_0 \frac{\boldsymbol{H}}{H}\right) - \frac{\tau_s}{I}\,\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-35}$$

50.1 Context

Section III of the paper opens by giving up the linear law (SM-15) and returning to the full equation (SM-13), now written with the Langevin magnetisation $M_0 = nmL[\xi(t)]$ in place of χH. Equation (SM-35) is therefore not a new equation — it is (SM-13) with the equilibrium term written out for arbitrary field strength. Its content is entirely in what it *no longer* assumes.

Three consequences follow immediately, and they organise the whole of this volume:

1. The equation is nonlinear in $\boldsymbol{H}$ through $L(\xi)$, so the circular-polarisation decomposition of Volume IV — which relied on superposition — is unavailable.

2. The equation is nonlinear in $\boldsymbol{M}$ through the last term, so even at fixed field the two magnetisation components couple.

3. A general analytic solution is therefore out of reach, and the authors proceed by a change of variables, a judicious choice of waveform, and finally numerics.

50.2 Symbols

Symbol	Meaning	Type	Units
$M_0 = nmL[\xi(t)]$	magnitude of the instantaneous-equilibrium magnetisation	scalar	G
$\boldsymbol{H}/H$	unit vector along the instantaneous field	vector	—
ξ	**in Sec. III**: the time-independent dimensionless field strength, $\xi = mH_0/kT$	scalar	—
$\gamma(t)$	periodic waveform of unit amplitude, $\xi(t) = \xi\gamma(t)$	function	—
$F(t)$	dimensionless longitudinal magnetisation, $M_x = nmF$	function	—
$G(t)$	dimensionless transverse magnetisation, $M_y = nm\Omega\tau_B G$	function	—
$\varepsilon = \pi/2\omega\tau_B$	dimensionless half-period, used from Ch. 54	scalar	—

Remark 50.2.1 (The third meaning of ξ). Volume IV warned that Sec. II of the paper uses the acting-field parameter $\xi = mH_0/kT\sqrt{2}$. Section III reverts to the amplitude-based $\xi = mH_0/kT$, which is also the ξ_0 of Bacri *et al.* The check is Sec. 54.7: at zero frequency (SM-40) must reduce to (SM-3) with the *static* Langevin parameter, and it does so only with the Sec. III convention. From this volume onwards, $\xi \equiv mH_0/kT$ unless explicitly stated otherwise.

50.3 The Waveform Factorisation

The authors write the time-dependent Langevin argument as

$$\xi(t) = \xi\,\gamma(t), \tag{50.3.1}$$

splitting it into a constant *strength* ξ and a dimensionless *waveform* $\gamma(t)$ of unit amplitude. For the harmonic field (SM-14), $\gamma(t) = \cos\omega t$; for the rectangular field used from Chapter 54 onwards, $\gamma(t) = \pm 1$ alternately.

The factorisation is useful because the two dependences play different roles: ξ controls how nonlinear the Langevin function is, while γ controls the time structure. Separating them makes it possible to change the waveform without touching the nonlinearity.

A property of L that makes the factorisation clean. The Langevin function is odd,

$$L(-x) = \coth(-x) - \frac{1}{-x} = -\coth x + \frac{1}{x} = -L(x), \tag{50.3.2}$$

using $\coth(-x) = -\coth x$ (both sinh and cosh have definite parity: sinh odd, cosh even). Consequently

$$M_0\frac{\boldsymbol{H}}{H} = nmL\big(\xi|\gamma|\big)\,\mathrm{sign}\big(\gamma\big)\hat{\boldsymbol{x}} = nmL\big(\xi\gamma\big)\hat{\boldsymbol{x}}, \tag{50.3.3}$$

so the awkward pair (magnitude, unit vector) collapses into the single signed expression $nmL(\xi\gamma)$. This is why the paper can write $M_0 = nmL[\xi(t)]$ without further comment even when the field reverses.

50.4 The Two-Component Decomposition

With the field along x and the vorticity along z, the magnetisation has only x and y components (nothing generates a z component: the torques lie in the xy plane's normal direction and the equations preserve $M_z = 0$). The authors introduce

$$\boxed{M_x = nm\,F(t), \qquad M_y = nm\,\Omega\tau_B\,G(t)} \tag{50.4.1}$$

Two things are built into this definition and both deserve comment.

(i) Why F and G are dimensionless. Both are magnetisations divided by the saturation value $nm = \varphi M_s$, so $|F| \le 1$ automatically. G carries an extra division by $\Omega\tau_B$.

(ii) Why the factor $\Omega\tau_B$ is extracted from M_y. Because M_y vanishes identically in a quiescent fluid — established in Sec. 44.6 for the harmonic case, and true generally by the symmetry argument of Sec. 27.5. The transverse magnetisation is therefore proportional to Ω for slow flows, and extracting that factor leaves G finite and Ω-independent as $\Omega \to 0$. The authors state the same thing as: the ratio M_y/M_x is of order $\Omega\tau_B$, since in a quiescent fluid only the x component survives.

Assumption 50.4.1 ($\Omega\tau_B$ is now the only small parameter)**.** Everything is computed to first order in $\Omega\tau_B$ and to *all* orders in ξ. This is the exact reverse of Volume V, where ξ was small and the frequency arbitrary.

50.5 Connection to the Next Equation

Substituting (50.4.1) into (SM-35) and separating components converts one nonlinear vector equation into two coupled scalar ones. Those are (SM-36).

Chapter 51

Equation (SM-36): The Coupled Equations for F and G

$$\begin{aligned}\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} &= L\big[\xi\gamma(t)\big] - F(t),\\ \tau_B \frac{\mathrm{d}G}{\mathrm{d}t} &= F(t) - \big(1 + \tfrac{1}{2}\,\xi\gamma(t)F(t)\big)\,G(t).\end{aligned} \tag{SM-36}$$

51.1 Context

This pair is the exact content of (SM-35) for the geometry at hand, to first order in $\Omega\tau_B$. Everything remaining in Sec. III of the paper is the solution of these two equations for various waveforms γ.

51.2 Step-by-Step Derivation

Step 1: the convective term. With $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\boldsymbol{\Omega} \times \boldsymbol{M} = \Omega\hat{\boldsymbol{z}} \times \big(M_x, M_y, 0\big) = \Omega\big(-M_y,\ M_x,\ 0\big), \tag{51.2.1}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$.

Step 2: the relaxation term. By (50.3.3),

$$-\frac{1}{\tau_B}\left(\boldsymbol{M} - M_0\frac{\boldsymbol{H}}{H}\right) = -\frac{1}{\tau_B}\Big(M_x - nmL(\xi\gamma),\ M_y,\ 0\Big). \tag{51.2.2}$$

Step 3: the nonlinear term. Expand the double cross product with the identity (1.7.1):

$$\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big) = \boldsymbol{M}\big(\boldsymbol{M} \cdot \boldsymbol{H}\big) - \boldsymbol{H}\big(\boldsymbol{M} \cdot \boldsymbol{M}\big) \tag{51.2.3}$$

$$= \big(M_x, M_y, 0\big)\, M_x H - \big(H, 0, 0\big)\big(M_x^2 + M_y^2\big), \tag{51.2.4}$$

since $\boldsymbol{H} = (H, 0, 0)$ gives $\boldsymbol{M} \cdot \boldsymbol{H} = M_x H$. Componentwise:

$$x: \qquad M_x^2 H - H\big(M_x^2 + M_y^2\big) = -H M_y^2, \tag{51.2.5}$$

$$y: \qquad M_y M_x H - 0 = M_x M_y H. \tag{51.2.6}$$

The x component is proportional to $M_y^2 = O\big((\Omega\tau_B)^2\big)$ and is therefore *second order*: it is dropped by Assumption 50.4.1. The y component is first order and must be kept.

Step 4: the x equation. Collecting the x components of (51.2.1), (51.2.2) and (51.2.5):

$$\frac{\mathrm{d}M_x}{\mathrm{d}t} = \underbrace{-\Omega M_y}_{O(\Omega^2\tau_B)} - \frac{M_x - nmL(\xi\gamma)}{\tau_B} + \underbrace{\frac{\tau_s}{I} H M_y^2}_{O(\Omega^2)}. \tag{51.2.7}$$

Both indicated terms are second order in $\Omega\tau_B$ and are dropped. Multiplying by τ_B and substituting $M_x = nmF$ from (50.4.1), the common factor nm cancels:

$$\boxed{\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = L\big[\xi\gamma(t)\big] - F(t)} \tag{51.2.8}$$

which is the first line of (SM-36). ✓ Note that it does not involve G at all: the longitudinal magnetisation is unaffected, at this order, by the flow. It can therefore be solved first, independently.

Step 5: the y equation. Collecting the y components:

$$\frac{\mathrm{d}M_y}{\mathrm{d}t} = \Omega M_x - \frac{M_y}{\tau_B} - \frac{\tau_s}{I} M_x M_y H. \tag{51.2.9}$$

Substitute $M_x = nmF$, $M_y = nm\Omega\tau_B G$ and $\tau_s/I = 1/6\eta\varphi$ [Eq. (7.5.12)]:

$$nm\Omega\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = \Omega\, nmF - \frac{nm\Omega\tau_B G}{\tau_B} - \frac{\big(nmF\big)\big(nm\Omega\tau_B G\big)H}{6\eta\varphi}. \tag{51.2.10}$$

Divide throughout by $nm\Omega$ (both nonzero):

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - G - \frac{nm\,\tau_B\,H}{6\eta\varphi}\,FG. \tag{51.2.11}$$

Step 6: evaluate the nonlinear coefficient. This is the same kind of cancellation met in Sec. 33.4. Insert $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$\frac{nm\tau_B H}{6\eta\varphi} = \frac{n\,m}{6\eta\,(nV)} \cdot \frac{3\eta V}{kT} \cdot H \tag{51.2.12}$$

$$= \frac{3\,n\,m\,\eta\,V\,H}{6\,\eta\,n\,V\,kT} \qquad \text{(collecting factors)} \tag{51.2.13}$$

$$= \frac{mH}{2kT} \qquad (n,\ V,\ \eta \text{ cancel; } 3/6 = 1/2) \tag{51.2.14}$$

$$= \frac{\xi\gamma(t)}{2}, \qquad \text{since } \frac{mH(t)}{kT} = \xi\gamma(t) \tag{51.2.15}$$

which is dimensionless, as it must be, and depends on time only through the waveform.

Step 7: assemble. Substituting (51.2.15) into (51.2.11):

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - G - \tfrac{1}{2}\xi\gamma FG \tag{51.2.16}$$

$$= F - \Big(1 + \tfrac{1}{2}\xi\gamma(t)F(t)\Big)G(t), \tag{51.2.17}$$

having factored G out of the last two terms. This is the second line of (SM-36). ■

51.3 Physical Interpretation

The F equation is pure Debye relaxation with a nonlinear drive. Its left side is the relaxation of the longitudinal magnetisation towards its instantaneous equilibrium $L[\xi\gamma(t)]$. Only the *drive* is nonlinear: the response is a simple exponential filter with time constant τ_B. At small ξ, $L \to \xi\gamma/3$ and the equation reduces to the linear relaxation equation (44.4.2) of Volume V.

The G equation contains the feedback. Compare the two equations: the F equation has the constant relaxation rate $1/\tau_B$, the G equation has the *enhanced* rate $(1 + \frac{1}{2}\xi\gamma F)/\tau_B$. The extra piece is the magnetic torque acting on the particles: a strongly magnetised grain in a strong field is mechanically pulled back into alignment, which destroys the transverse magnetisation faster than Brownian motion alone would. It is the same feedback that produced the denominator $1 + \zeta$ in Eq. (15.5.17), and indeed $\frac{1}{2}\xi\gamma F$ is the instantaneous version of $\zeta = \frac{1}{2}\xi L(\xi)$: in a stationary field $F \to L(\xi)$ and $\gamma = 1$, whereupon $\frac{1}{2}\xi\gamma F \to \frac{1}{2}\xi L(\xi) = \zeta$ exactly. ✓ This is a strong internal consistency check linking Volume II to Volume VI.

The drive of G is F. The transverse magnetisation is generated by the vortex sweeping the longitudinal one sideways. No F, no G; no Ω, no G either (that factor was extracted in (50.4.1)).

51.4 Connection to the Next Equation

The quantity we actually need is not G itself but the averaged torque, which involves $\overline{G\gamma}$. That is (SM-37).

Chapter 52

Equation (SM-37): The Averaged Torque at Arbitrary Amplitude

$$\overline{\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z} = -\overline{M_y H_x} = -3\eta\varphi\,\Omega\,\xi\,\overline{G(t)\gamma(t)}. \tag{SM-37}$$

52.1 Context

Volume V established that the flow responds only to the period average of the torque (Chapter 40). That conclusion used nothing about the field amplitude, so it carries over unchanged. What changes is the value of the average, which must now be expressed through the unknown function G.

52.2 Step-by-Step Derivation

Step 1: the torque has only a z component. Exactly as in Sec. 39.4, Step 1: with $\boldsymbol{H} = (H, 0, 0)$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z = M_x H_y - M_y H_x = -M_y H_x. \tag{52.2.1}$$

Step 2: substitute the definitions. With $M_y = nm\Omega\tau_B G(t)$ from (50.4.1) and $H_x = H_0\gamma(t)$,

$$\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z = -nm\,\Omega\,\tau_B\,H_0\,G(t)\gamma(t). \tag{52.2.2}$$

Step 3: average over the period. The factors nm, Ω, τ_B and H_0 are constants on the fast time scale, so

$$\overline{\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z} = -nm\,\Omega\,\tau_B\,H_0\,\overline{G\gamma}. \tag{52.2.3}$$

Step 4: evaluate the prefactor. Insert $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$nm\tau_B H_0 = n\,m\,\frac{3\eta V}{kT}\,H_0 \tag{52.2.4}$$

$$= 3\eta\,(nV)\,\frac{mH_0}{kT} \qquad \text{(regrouping)} \tag{52.2.5}$$

$$= 3\eta\varphi\,\xi, \qquad \text{since } \xi = \frac{mH_0}{kT} \text{ in Sec. III} \tag{52.2.6}$$

Step 5: assemble. Substituting (52.2.6) into (52.2.3):

$$\overline{\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z} = -3\eta\varphi\Omega\xi\,\overline{G\gamma}, \tag{SM-37}$$

■

52.3 Physical Interpretation

The whole of the finite-amplitude problem has now been compressed into the single number $\overline{G\gamma}$: the correlation between the transverse magnetisation and the driving waveform. If G is in phase with γ the correlation is positive, the torque opposes the vorticity, and the viscosity rises. If G is predominantly out of phase — which is what a large lag does — the correlation becomes negative and so does the viscosity. Everything hinges on a phase relationship, exactly as in the weak-field theory, where the same role was played by $\cos 2\alpha$.

52.4 Connection to the Next Equation

Converting the averaged torque into a viscosity gives (SM-38), and the average itself is defined in (SM-39).

Chapter 53

Equations (SM-38) and (SM-39): The General Viscosity Formula

$$\Delta\eta = \tfrac{3}{4}\,\eta\varphi\,\xi\,\overline{G(t)\gamma(t)}, \tag{SM-38}$$

$$\overline{G(t)\gamma(t)} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} G(t)\gamma(t)\,\mathrm{d}t. \tag{SM-39}$$

53.1 Context

Equation (SM-39) is simply the definition of the period average, restated for this particular product. Equation (SM-38) converts the torque into a viscosity by the route of Chapter 41.

Remark 53.1.1 (On the numerical coefficient). The scanned text of the paper renders the coefficient of (SM-38) indistinctly. Its value is fixed unambiguously by internal consistency: (SM-38) must, when combined with the relation $(\xi/2)\overline{G\gamma} = (\Omega - \Theta)/\Omega$ derived in Sec. 53.4 below, reproduce the master formula (SM-34). That requirement gives $\frac{3}{4}$, and the derivation of Sec. 53.3 obtains $\frac{3}{4}$ independently from the momentum equation. We therefore use $\frac{3}{4}$ throughout.

53.2 Symbols

Nothing new. Note that $\Delta\eta$ here depends on ξ both explicitly and through G, which is why the dependence on field strength turns out to be non-monotonic (Sec. 57.3).

53.3 Step-by-Step Derivation of (SM-38)

Step 1: the magnetic force density. By the averaged momentum equation (SM-26), the magnetic contribution is $\frac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}}$. Writing the averaged torque of (SM-37) as a vector — it points along $\hat{\boldsymbol{z}}$, and $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$, so

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -3\eta\varphi\,\xi\,\overline{G\gamma}\;\boldsymbol{\Omega}, \tag{53.3.1}$$

we obtain

$$\tfrac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}} = -\tfrac{3}{2}\eta\varphi\xi\,\overline{G\gamma}\;\nabla\times\boldsymbol{\Omega}. \tag{53.3.2}$$

Step 2: add the ordinary viscous term. By Eq. (41.2.6), $\eta\Delta\boldsymbol{v} = -2\eta\nabla\times\boldsymbol{\Omega}$. Hence

$$\eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}} = -2\eta\nabla\times\boldsymbol{\Omega} - \tfrac{3}{2}\eta\varphi\xi\overline{G\gamma}\,\nabla\times\boldsymbol{\Omega} \tag{53.3.3}$$

$$= -2\Big(\eta + \tfrac{3}{4}\eta\varphi\xi\overline{G\gamma}\Big)\nabla\times\boldsymbol{\Omega}. \tag{53.3.4}$$

Step 3: read off the viscosity.

$$\Delta\eta = \tfrac{3}{4}\eta\varphi\xi\,\overline{G(t)\gamma(t)}, \tag{SM-38}$$

■

53.4 Consistency: Recovering (SM-34)

The authors verify their formula by comparing it with the torque balance, and we follow them.

Step 1. From Eq. (14.2.1), the balance of magnetic and viscous torques gives $\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega}-\boldsymbol{\Theta})$.

Step 2. Compare with (53.3.1):

$$-6\eta\varphi\big(\boldsymbol{\Omega}-\boldsymbol{\Theta}\big) = -3\eta\varphi\xi\overline{G\gamma}\,\boldsymbol{\Omega}. \tag{53.4.1}$$

Dividing by $-6\eta\varphi\,\Omega$:

$$\boxed{\frac{\Omega-\Theta}{\Omega} = \frac{\xi}{2}\,\overline{G(t)\gamma(t)}} \tag{53.4.2}$$

which is exactly the relation the authors state between their Eqs. (SM-39) and (SM-40).

Step 3. Substitute (53.4.2) into (SM-38):

$$\Delta\eta = \tfrac{3}{4}\eta\varphi\xi\overline{G\gamma} = \tfrac{3}{4}\eta\varphi\cdot 2\cdot\frac{\Omega-\Theta}{\Omega} = \tfrac{3}{2}\eta\varphi\,\frac{\Omega-\Theta}{\Omega}, \tag{SM-34}$$

recovering the master formula. ✓

What this establishes. Equation (SM-34) was derived in Volume II from angular-momentum bookkeeping, and in Volume V from the explicit weak-field solution. Here it is recovered from the *finite-amplitude* theory. The authors draw the correct conclusion: a formula obtained under the weak-field assumption in fact holds for any field strength and frequency, because it never used the weak-field assumption in the first place — only the torque balance.

53.5 Connection to the Next Equation

To get numbers we must solve (SM-36). The authors do so first analytically, in the weak-field limit and for a rectangular waveform. That is (SM-40).

Chapter 54

Equation (SM-40): The Square-Wave Solution

$$\Delta\eta = \frac{1}{4}\,\eta\varphi\,\xi^2\left(2 - \tanh^2\varepsilon - 2\,\frac{\tanh\varepsilon}{\varepsilon}\right), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}. \tag{SM-40}$$

54.1 Context

The authors note that Eqs. (SM-36) can be solved relatively easily if the field varies not harmonically but in a stepwise way — that is, if $\gamma(t)$ is a rectangular periodic function with period $T_0 = 2\pi/\omega$, as drawn in their Fig. 2. In weak fields the nonlinear term in the second equation may then be neglected and the problem solved in closed form.

The result is important for two reasons. First, it is the only closed-form finite-frequency result in Sec. III. Second, comparing it with the harmonic result (SM-28) shows how little the answer depends on the waveform — which is what licenses the use of a square wave for the numerics that follow.

Remark 54.1.1 (A misprint in the published equation). The published (SM-40) has $\tanh\varepsilon$ where the derivation below gives $\tanh^2\varepsilon$. That the square belongs there is confirmed by the paper's own numbers: with $\tanh^2\varepsilon$ the sign change occurs at $\omega\tau_B = 0.978$ and the minimum at $\omega\tau_B = 1.707$ with depth $-0.105\,\Delta\eta(0)$, which are precisely the three values the authors quote in the following paragraph. We verify this arithmetic explicitly in Sec. 54.7.

54.2 Symbols

Symbol	Meaning	Type	Units
$\gamma(t)$	rectangular waveform: $+1$ for $0 < t < \pi/\omega$, -1 for $\pi/\omega < t < 2\pi/\omega$	function	—
$T_h = \pi/\omega$	half-period of the field	scalar	s
$\varepsilon = T_h/2\tau_B = \pi/2\omega\tau_B$	half-period measured in units of $2\tau_B$	dimensionless	—
$u = e^{-2\varepsilon}$	abbreviation used in the algebra	dimensionless	—

Note the inverse relation between ε and frequency: $\varepsilon \to \infty$ is the static limit, $\varepsilon \to 0$ the

high-frequency limit.

54.3 Assumptions

1. Weak field, $\xi \ll 1$, so that $L(\xi\gamma) \simeq \xi\gamma/3$ [Eq. (10.3.8)] and the nonlinear term $\frac{1}{2}\xi\gamma FG$ in the second of (SM-36) — which carries an explicit factor ξ and is therefore of higher order — may be dropped.
2. Rectangular waveform.
3. Steady periodic state: all transients have died.
4. $\Omega\tau_B \ll 1$, as always.

54.4 Mathematical Background: antiperiodicity

Result 54.4.1 (Half-period antisymmetry)**.** *In the steady state, for a waveform satisfying* $\gamma(t+T_h) = -\gamma(t)$*,*

$$F\big(t+T_h\big) = -F(t), \qquad G\big(t+T_h\big) = -G(t). \tag{54.4.1}$$

Derivation. Consider the transformation $t \to t+T_h$, $F \to -F$, $G \to -G$. Under it, the first of (SM-36) becomes

$$\tau_B \frac{\mathrm{d}(-F)}{\mathrm{d}t} = L\big[\xi\gamma(t+T_h)\big] - (-F) = L\big[-\xi\gamma(t)\big] + F = -L\big[\xi\gamma(t)\big] + F, \tag{54.4.2}$$

using the oddness (50.3.2); multiplying by -1 returns the original equation. The second of (SM-36) is likewise invariant: the term $\frac{1}{2}\xi\gamma F$ is even under the transformation (both γ and F change sign), so $(1+\frac{1}{2}\xi\gamma F)G \to -(1+\frac{1}{2}\xi\gamma F)G$, matching the sign change of F on the right and of $\mathrm{d}G/\mathrm{d}t$ on the left. The transformed functions therefore solve the same equations; and since the steady periodic solution is unique (the equations are linear-in-G relaxation equations with a unique attracting periodic orbit), the solution must map to itself, giving (54.4.1). ■

This result does the work of halving the problem: it suffices to solve on $0 < t < T_h$, where $\gamma = +1$, and impose (54.4.1) as the boundary condition instead of an initial condition.

54.5 Step-by-Step Derivation

Throughout, $0 < t < T_h$ and $\gamma = +1$, and we write $u \equiv e^{-2\varepsilon} = e^{-T_h/\tau_B}$ — the two agree because $T_h/\tau_B = 2\varepsilon$ by the definition of ε.

Step 1: solve for F. With $L(\xi\gamma) \simeq \xi/3$ the first of (SM-36) is

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} + F = \frac{\xi}{3}, \tag{54.5.1}$$

a linear equation with constant coefficients. Its general solution is the particular solution $\xi/3$ plus the decaying homogeneous solution:

$$F(t) = \frac{\xi}{3} + \left(F_0 - \frac{\xi}{3}\right) e^{-t/\tau_B}, \qquad F_0 \equiv F(0). \tag{54.5.2}$$

Step 2: impose antiperiodicity on F. Setting $t = T_h$ in (54.5.2) and using $F(T_h) = -F_0$:

$$-F_0 = \frac{\xi}{3} + \left(F_0 - \frac{\xi}{3}\right) u \tag{54.5.3}$$

$$-F_0 - F_0 u = \frac{\xi}{3} - \frac{\xi}{3} u \tag{54.5.4}$$

$$-F_0\big(1+u\big) = \frac{\xi}{3}\big(1-u\big) \tag{54.5.5}$$

$$F_0 = -\frac{\xi}{3}\,\frac{1-u}{1+u} = -\frac{\xi}{3}\tanh\varepsilon, \tag{54.5.6}$$

the last step using

$$\tanh\varepsilon = \frac{e^{\varepsilon} - e^{-\varepsilon}}{e^{\varepsilon} + e^{-\varepsilon}} = \frac{1 - e^{-2\varepsilon}}{1 + e^{-2\varepsilon}} = \frac{1-u}{1+u}, \tag{54.5.7}$$

obtained by multiplying numerator and denominator by $e^{-\varepsilon}$.

Step 3: the explicit F. Substituting (54.5.6) into (54.5.2):

$$F(t) = \frac{\xi}{3} + \left(-\frac{\xi}{3}\tanh\varepsilon - \frac{\xi}{3}\right) e^{-t/\tau_B} \tag{54.5.8}$$

$$= \frac{\xi}{3}\Big[1 - A\,e^{-t/\tau_B}\Big], \qquad A \equiv 1 + \tanh\varepsilon = \frac{2}{1+u}, \tag{54.5.9}$$

where the value of A follows from (54.5.7): $1 + (1-u)/(1+u) = \big[(1+u)+(1-u)\big]/(1+u) = 2/(1+u)$.

Step 4: the equation for G. Dropping the nonlinear term (Assumption 1), the second of (SM-36) is

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} + G = F(t) = \frac{\xi}{3}\Big[1 - A e^{-t/\tau_B}\Big]. \tag{54.5.10}$$

Step 5: solve it. The right-hand side has two pieces, and we find a particular solution for each.

(a) The constant $\xi/3$. A constant $G = \xi/3$ satisfies $\tau_B \cdot 0 + \xi/3 = \xi/3$. ✓

(b) The term $-(\xi A/3)e^{-t/\tau_B}$. This is *resonant*: e^{-t/τ_B} is itself a solution of the homogeneous equation, so the usual guess Be^{-t/τ_B} fails. The standard remedy is to multiply by t; try $G = Bte^{-t/\tau_B}$. Then

$$\tau_B \frac{\mathrm{d}}{\mathrm{d}t}\Big(Bte^{-t/\tau_B}\Big) + Bte^{-t/\tau_B} = \tau_B B\left(e^{-t/\tau_B} - \frac{t}{\tau_B}e^{-t/\tau_B}\right) + Bte^{-t/\tau_B} \tag{54.5.11}$$

$$= \tau_B B e^{-t/\tau_B} - Bte^{-t/\tau_B} + Bte^{-t/\tau_B} \tag{54.5.12}$$

$$= \tau_B B e^{-t/\tau_B}, \tag{54.5.13}$$

so matching to $-(\xi A/3)e^{-t/\tau_B}$ requires

$$B = -\frac{\xi A}{3\tau_B}. \tag{54.5.14}$$

(c) Assemble. Adding the homogeneous solution Ce^{-t/τ_B}:

$$G(t) = \frac{\xi}{3} - \frac{\xi A}{3\tau_B}\,t\,e^{-t/\tau_B} + C\,e^{-t/\tau_B}. \tag{54.5.15}$$

Step 6: impose antiperiodicity on G. At $t = T_h = 2\varepsilon\tau_B$,

$$G(T_h) = \frac{\xi}{3} - \frac{\xi A}{3\tau_B}\big(2\varepsilon\tau_B\big)u + Cu = \frac{\xi}{3} - \frac{2\xi A\varepsilon}{3}u + Cu, \tag{54.5.16}$$

and this must equal $-G(0) = -\big(\xi/3 + C\big)$. Hence

$$\frac{\xi}{3} - \frac{2\xi A\varepsilon}{3}u + Cu = -\frac{\xi}{3} - C \tag{54.5.17}$$

$$C\big(1+u\big) = -\frac{2\xi}{3} + \frac{2\xi A\varepsilon}{3}u \tag{54.5.18}$$

$$C = \frac{2\xi}{3}\cdot\frac{A\varepsilon u - 1}{1+u}. \tag{54.5.19}$$

Step 7: the average we need. By (SM-41), proved in Chapter 55, the average of $G\gamma$ over the full period equals the average of G over the half period on which $\gamma = +1$:

$$\overline{G\gamma} = \frac{1}{T_h}\int_0^{T_h} G(t)\,\mathrm{d}t. \tag{54.5.20}$$

Step 8: perform the integral. Three elementary integrals are needed:

$$\int_0^{T_h} \mathrm{d}t = T_h = 2\varepsilon\tau_B, \tag{54.5.21}$$

$$\int_0^{T_h} e^{-t/\tau_B}\mathrm{d}t = \tau_B\big(1-u\big), \tag{54.5.22}$$

$$\int_0^{T_h} t\,e^{-t/\tau_B}\mathrm{d}t = \tau_B^2\Big[1 - \big(1+2\varepsilon\big)u\Big]. \tag{54.5.23}$$

The last one is obtained by parts: with $\mathrm{d}v = e^{-t/\tau_B}\mathrm{d}t$, $v = -\tau_B e^{-t/\tau_B}$,

$$\int_0^{T} te^{-t/\tau_B}\mathrm{d}t = \Big[-\tau_B t e^{-t/\tau_B}\Big]_0^{T} + \tau_B\int_0^{T} e^{-t/\tau_B}\mathrm{d}t \tag{54.5.24}$$

$$= -\tau_B T e^{-T/\tau_B} + \tau_B^2\Big(1 - e^{-T/\tau_B}\Big), \tag{54.5.25}$$

and putting $T = T_h = 2\varepsilon\tau_B$, $e^{-T/\tau_B} = u$ gives $-2\varepsilon\tau_B^2 u + \tau_B^2(1-u) = \tau_B^2[1-(1+2\varepsilon)u]$. ✓

Therefore

$$\int_0^{T_h} G\,\mathrm{d}t = \frac{\xi}{3}\big(2\varepsilon\tau_B\big) - \frac{\xi A}{3\tau_B}\,\tau_B^2\Big[1-\big(1+2\varepsilon\big)u\Big] + C\tau_B\big(1-u\big). \tag{54.5.26}$$

Dividing by $T_h = 2\varepsilon\tau_B$:

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{\xi A}{6\varepsilon}\Big[1-\big(1+2\varepsilon\big)u\Big] + \frac{C\big(1-u\big)}{2\varepsilon}. \tag{54.5.27}$$

Step 9: simplify the second term. With $A = 2/(1+u)$,

$$-\frac{\xi A}{6\varepsilon}\Big[1-\big(1+2\varepsilon\big)u\Big] = -\frac{\xi}{3\varepsilon}\cdot\frac{1-u-2\varepsilon u}{1+u} \tag{54.5.28}$$

$$= -\frac{\xi}{3\varepsilon}\cdot\frac{1-u}{1+u} + \frac{\xi}{3\varepsilon}\cdot\frac{2\varepsilon u}{1+u} \tag{54.5.29}$$

$$= -\frac{\xi\tanh\varepsilon}{3\varepsilon} + \frac{2\xi u}{3\big(1+u\big)}, \tag{54.5.30}$$

using (54.5.7) in the last line.

Step 10: simplify the third term. With (54.5.19),

$$\frac{C(1-u)}{2\varepsilon} = \frac{2\xi}{3}\cdot\frac{A\varepsilon u - 1}{1+u}\cdot\frac{1-u}{2\varepsilon} \tag{54.5.31}$$

$$= \frac{\xi}{3\varepsilon}(A\varepsilon u - 1)\frac{1-u}{1+u} \tag{54.5.32}$$

$$= \frac{\xi}{3\varepsilon}(A\varepsilon u - 1)\tanh\varepsilon \tag{54.5.33}$$

$$= \frac{\xi u A\tanh\varepsilon}{3} - \frac{\xi\tanh\varepsilon}{3\varepsilon}. \tag{54.5.34}$$

For the first piece, use $A\tanh\varepsilon = \dfrac{2}{1+u}\cdot\dfrac{1-u}{1+u} = \dfrac{2(1-u)}{(1+u)^2}$:

$$\frac{\xi u A\tanh\varepsilon}{3} = \frac{2\xi u(1-u)}{3(1+u)^2}. \tag{54.5.35}$$

Step 11: collect. Substituting (54.5.30), (54.5.34) and (54.5.35) into (54.5.27):

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{2\xi\tanh\varepsilon}{3\varepsilon} + \underbrace{\frac{2\xi u}{3(1+u)} + \frac{2\xi u(1-u)}{3(1+u)^2}}_{\equiv S}. \tag{54.5.36}$$

Step 12: simplify S. Put the two terms over the common denominator $(1+u)^2$:

$$S = \frac{2\xi u}{3}\left[\frac{1}{1+u} + \frac{1-u}{(1+u)^2}\right] = \frac{2\xi u}{3}\cdot\frac{(1+u)+(1-u)}{(1+u)^2} = \frac{2\xi u}{3}\cdot\frac{2}{(1+u)^2} = \frac{4\xi u}{3(1+u)^2}. \tag{54.5.37}$$

Now note the identity

$$1-\tanh^2\varepsilon = 1 - \frac{(1-u)^2}{(1+u)^2} = \frac{(1+u)^2-(1-u)^2}{(1+u)^2} = \frac{4u}{(1+u)^2}, \tag{54.5.38}$$

the numerator having been expanded as $(1+2u+u^2)-(1-2u+u^2) = 4u$. Hence

$$S = \frac{\xi}{3}\left(1-\tanh^2\varepsilon\right). \tag{54.5.39}$$

Step 13: the average. Substituting into (54.5.36):

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{2\xi\tanh\varepsilon}{3\varepsilon} + \frac{\xi}{3}\left(1-\tanh^2\varepsilon\right) \tag{54.5.40}$$

$$= \frac{\xi}{3}\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right). \tag{54.5.41}$$

Step 14: the viscosity. Insert (54.5.41) into (SM-38):

$$\Delta\eta = \frac{3}{4}\eta\varphi\xi\cdot\frac{\xi}{3}\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right) \tag{54.5.42}$$

$$= \frac{1}{4}\eta\varphi\xi^2\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right), \tag{SM-40}$$

since $3/(4\cdot 3) = 1/4$. ■

54.6 Consistency Checks and Verification Against the Paper's Numbers

Write

$$f(\varepsilon) \equiv 2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}, \qquad \Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 f(\varepsilon). \tag{54.6.1}$$

(a) The static limit $\varepsilon \to \infty$ (i.e. $\omega \to 0$). Then $\tanh\varepsilon \to 1$ and $\tanh\varepsilon/\varepsilon \to 0$, so

$$f(\infty) = 2 - 1 - 0 = 1, \qquad \Delta\eta \to \tfrac{1}{4}\eta\varphi\xi^2, \tag{54.6.2}$$

which is exactly (SM-3), the stationary weak-field result. ✓ Note that here $\xi = mH_0/kT$ with no factor $\sqrt{2}$: a square wave at zero frequency is a *constant* field of magnitude H_0, not an oscillating one, so no root-mean-square correction arises. This is the promised confirmation that Sec. III uses the amplitude-based convention.

(b) The high-frequency limit $\varepsilon \to 0$. Use $\tanh\varepsilon = \varepsilon - \varepsilon^3/3 + O(\varepsilon^5)$, whence $\tanh^2\varepsilon = \varepsilon^2 + O(\varepsilon^4)$ and $\tanh\varepsilon/\varepsilon = 1 - \varepsilon^2/3 + O(\varepsilon^4)$. Then

$$f(\varepsilon) = 2 - \varepsilon^2 - 2\left(1 - \frac{\varepsilon^2}{3}\right) + O(\varepsilon^4) \tag{54.6.3}$$

$$= 2 - \varepsilon^2 - 2 + \frac{2\varepsilon^2}{3} \tag{54.6.4}$$

$$= -\frac{\varepsilon^2}{3} + O(\varepsilon^4). \tag{54.6.5}$$

So $\Delta\eta \to -\frac{1}{12}\eta\varphi\xi^2\varepsilon^2 < 0$, and since $\varepsilon \propto 1/\omega$ this is again the ω^{-2} decay of (SM-29). ✓

(c) The sign change. Solve $f(\varepsilon_0) = 0$ numerically:

ε	$\tanh\varepsilon$	$\tanh^2\varepsilon$	$2\tanh\varepsilon/\varepsilon$	$f(\varepsilon)$
1.55	0.9138	0.8350	1.1791	−0.0141
1.60	0.9217	0.8495	1.1521	−0.0016
1.61	0.9232	0.8523	1.1468	+0.0009
1.70	0.9354	0.8750	1.1005	+0.0246

The root is $\varepsilon_0 \approx 1.606$, corresponding to

$$\omega_0\tau_B = \frac{\pi}{2\varepsilon_0} = \frac{3.1416}{3.212} = 0.978 \approx 0.98. \quad \checkmark \tag{54.6.6}$$

This is precisely the value quoted in the paper for the rectangular case.

(d) The minimum. Evaluating f near $\varepsilon \approx 0.9$:

ε	$\tanh\varepsilon$	$\tanh^2\varepsilon$	$2\tanh\varepsilon/\varepsilon$	$f(\varepsilon)$
0.85	0.6911	0.4776	1.6260	−0.1036
0.90	0.7163	0.5131	1.5918	−0.1049
0.92	0.7259	0.5269	1.5779	−0.1048
0.95	0.7398	0.5473	1.5574	−0.1047

The minimum is $f_{\min} = -0.1049$ at $\varepsilon_m \approx 0.920$, i.e.

$$\omega_m \tau_B = \frac{\pi}{2\varepsilon_m} = \frac{3.1416}{1.840} = 1.707, \qquad \frac{\Delta\eta_m}{\Delta\eta(0)} = \frac{f_{\min}}{f(\infty)} = -0.105. \quad \checkmark \tag{54.6.7}$$

Both agree with the values printed in the paper to the quoted precision. *This is the verification promised in Sec. 54.1: the reading* $\tanh^2 \varepsilon$ *reproduces all three of the authors' numbers, whereas the literal printed* $\tanh \varepsilon$ *does not.*

54.7 Comparison of the Two Waveforms

Quantity	Harmonic (SM-28)	Rectangular (SM-40)
sign change, $\omega_0 \tau_B$	1	0.98
minimum, $\omega_m \tau_B$	$\sqrt{3} = 1.732$	1.707
depth, $\Delta\eta_m / \Delta\eta(0)$	$-1/8 = -0.125$	-0.105
static value $\Delta\eta(0)$	$\frac{1}{4}\eta\varphi\xi^2$	$\frac{1}{4}\eta\varphi\xi^2$

The two curves are not merely qualitatively but *quantitatively* similar, as the authors observe. That is the licence for what follows: since the answer barely depends on the waveform, one may as well use the waveform for which the equations integrate — the square wave.

Why the square wave gives a slightly shallower minimum. A square wave holds the field at full strength throughout the cycle, whereas a cosine spends much of each cycle near zero. One might expect the square wave to be *more* effective. What matters, however, is not the mean strength but the phase correlation, and the abrupt reversals of the square wave excite a broad spectrum of harmonics, only the fundamental of which is optimally phased. The net result is a 16% shallower minimum.

54.8 Connection to the Next Equation

The half-period reduction (54.5.20) used in Step 7 is stated by the authors as (SM-41), and we prove it next.

Chapter 55

Equation (SM-41): The Half-Period Reduction

$$\overline{G(t)\gamma(t)} = \frac{\omega}{\pi}\int_0^{\pi/\omega} G(t)\,\mathrm{d}t. \tag{SM-41}$$

55.1 Context

The authors note that Eqs. (SM-36) imply $G(t) = -G(t+\pi/\omega)$, so that instead of (SM-39) one may use (SM-41): it is sufficient to know the transverse magnetisation over half a period only. This halves the numerical work and, in the analytic solution of Chapter 54, made the problem tractable at all.

55.2 Step-by-Step Derivation

Step 1: antiperiodicity. Proved as (54.4.1) in Sec. 54.4: $G(t+T_h) = -G(t)$ with $T_h = \pi/\omega$. The waveform obeys the same relation, $\gamma(t+T_h) = -\gamma(t)$, by construction.

Step 2: the product is periodic with the *half* period.

$$G\big(t+T_h\big)\gamma\big(t+T_h\big) = \big(-G(t)\big)\big(-\gamma(t)\big) = G(t)\gamma(t). \tag{55.2.1}$$

A product of two antiperiodic functions is periodic on the half period.

Step 3: the full-period average equals the half-period average. Split the integral in (SM-39) at $t = T_h$ and substitute $t = t' + T_h$ in the second piece:

$$\begin{aligned}
\int_0^{2T_h} G\gamma\,\mathrm{d}t &= \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_{T_h}^{2T_h} G\gamma\,\mathrm{d}t && (55.2.2)\\
&= \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_0^{T_h} G\big(t'+T_h\big)\gamma\big(t'+T_h\big)\,\mathrm{d}t' && (55.2.3)\\
&= \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_0^{T_h} G\gamma\,\mathrm{d}t' \qquad \text{by (55.2.1)} && (55.2.4)\\
&= 2\int_0^{T_h} G\gamma\,\mathrm{d}t. && (55.2.5)
\end{aligned}$$

Dividing by the full period $2T_h = 2\pi/\omega$:

$$\overline{G\gamma} = \frac{\omega}{2\pi} \cdot 2 \int_0^{T_h} G\gamma \, \mathrm{d}t = \frac{\omega}{\pi} \int_0^{\pi/\omega} G\gamma \, \mathrm{d}t. \tag{55.2.6}$$

Step 4: drop γ. On the first half period $\gamma = +1$ by construction, so $G\gamma = G$ there, and

$$\overline{G\gamma} = \frac{\omega}{\pi} \int_0^{\pi/\omega} G(t) \, \mathrm{d}t, \tag{SM-41}$$

■

Remark 55.2.1 (The reduction is not merely convenient). For a harmonic waveform $\gamma = \cos \omega t$ the same antiperiodicity holds ($\cos \omega(t + \pi/\omega) = -\cos \omega t$), so (SM-41) applies there too — except that γ cannot then be dropped from the integrand, since it is not constant on the half period. It is the *rectangular* waveform that permits Step 4, and this is the main computational reason the authors adopt it.

55.3 Connection to the Next Equation

At finite amplitude the equations must be integrated with the full Langevin function. The formal solution is (SM-42), with the auxiliary functions (SM-43).

Chapter 56

Equations (SM-42) and (SM-43): The Formal Solution

$$G(t) = \frac{e^{g(t)}}{\tau_B}\left(-\frac{e^{g(\pi/\omega)}}{1+e^{g(\pi/\omega)}}\int_0^{\pi/\omega} F(t)e^{-g(t)}\,\mathrm{d}t + \int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right), \tag{SM-42}$$

$$\begin{aligned} F(t) &= L(\xi)\left(1 - \frac{2e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}\right), \\ g(t) &= -\frac{t}{\tau_B}\left(1+\tfrac{1}{2}\xi L(\xi)\right) + \xi L(\xi)\,\frac{1-e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}. \end{aligned} \tag{SM-43}$$

56.1 Context

These are the finite-amplitude counterparts of Chapter 54. The nonlinear term is now retained, and the Langevin function is used in full. The result is not a closed-form $\Delta\eta$ but a closed-form *quadrature*: G is expressed through integrals of known functions, which can then be evaluated numerically. This is what produces Figs. 3–5 of the paper.

56.2 Assumptions

1. Rectangular waveform, $\gamma = +1$ on $0 < t < \pi/\omega$.
2. Steady periodic state, with the antiperiodicity (54.4.1).
3. $\Omega\tau_B \ll 1$; *no* restriction on ξ.

56.3 Derivation of $F(t)$

With $\gamma = +1$ the first of (SM-36) reads

$$\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} + F = L(\xi), \tag{56.3.1}$$

identical in structure to Step 1 of Sec. 54.5 but with $\xi/3$ replaced by the *exact* $L(\xi)$. Every step of Steps 1–3 there goes through unchanged with that substitution, giving

$$F(t) = L(\xi)\left[1 - \frac{2}{1+u}e^{-t/\tau_B}\right], \qquad u = e^{-\pi/\omega\tau_B}, \tag{56.3.2}$$

which is the first line of (SM-43). ✓ (Note that $u = e^{-T_h/\tau_B}$ and $T_h = \pi/\omega$, so $u = e^{-\pi/\omega\tau_B}$ exactly as printed.)

Sanity check. At weak field $L(\xi) \to \xi/3$ and (56.3.2) reduces to (54.5.9). ✓

56.4 Derivation of $g(t)$

Step 1: put the G equation in standard linear form. With $\gamma = +1$, the second of (SM-36) is

$$\frac{\mathrm{d}G}{\mathrm{d}t} + \frac{1 + \tfrac{1}{2}\xi F(t)}{\tau_B}\, G = \frac{F(t)}{\tau_B}. \tag{56.4.1}$$

This is a first-order linear equation with a *time-dependent* coefficient, because F depends on t.

Step 2: the integrating factor. For $\mathrm{d}G/\mathrm{d}t + P(t)G = Q(t)$ the integrating factor is $\exp\!\int P\,\mathrm{d}t$. Define

$$g(t) \equiv -\frac{1}{\tau_B}\int_0^t \Big(1 + \tfrac{1}{2}\xi F(t')\Big)\mathrm{d}t', \tag{56.4.2}$$

so that the integrating factor is $e^{-g(t)}$ and $g(0) = 0$. (The minus sign in the definition is the authors' convention; it makes g negative and decreasing, so that e^{g} is a decaying factor.)

Step 3: evaluate the integral. Insert (56.3.2):

$$\int_0^t F\,\mathrm{d}t' = L(\xi)\int_0^t \left[1 - \frac{2}{1+u}e^{-t'/\tau_B}\right]\mathrm{d}t' \tag{56.4.3}$$

$$= L(\xi)\left[t - \frac{2}{1+u}\cdot\tau_B\Big(1 - e^{-t/\tau_B}\Big)\right]. \tag{56.4.4}$$

Therefore

$$g(t) = -\frac{1}{\tau_B}\left[t + \frac{\xi}{2}L(\xi)\,t - \frac{\xi L(\xi)\tau_B\big(1 - e^{-t/\tau_B}\big)}{1+u}\right] \tag{56.4.5}$$

$$= -\frac{t}{\tau_B}\left(1 + \tfrac{1}{2}\xi L(\xi)\right) + \xi L(\xi)\,\frac{1 - e^{-t/\tau_B}}{1+u}, \tag{56.4.6}$$

which is the second line of (SM-43). ■

Reading g. The first term is a steady decay at the enhanced rate $\big(1 + \tfrac{1}{2}\xi L\big)/\tau_B$ — the Brownian rate augmented by the magnetic restoring torque, exactly the feedback identified in Sec. 51.3. The second term is a transient correction that accounts for F not having reached its plateau immediately after each reversal of the field.

56.5 Derivation of (SM-42)

Step 1: integrate the linear equation. Multiply (56.4.1) by $e^{-g(t)}$. By construction $\mathrm{d}(-g)/\mathrm{d}t = \big(1 + \tfrac{1}{2}\xi F\big)/\tau_B$, so the left-hand side is an exact derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(Ge^{-g}\Big) = e^{-g}\frac{\mathrm{d}G}{\mathrm{d}t} - g'e^{-g}G = e^{-g}\left[\frac{\mathrm{d}G}{\mathrm{d}t} + \frac{1 + \tfrac{1}{2}\xi F}{\tau_B}G\right] = \frac{F}{\tau_B}e^{-g}. \tag{56.5.1}$$

Step 2: integrate from 0 to t. With $g(0) = 0$,

$$G(t)e^{-g(t)} - G(0) = \frac{1}{\tau_B}\int_0^t F(t')e^{-g(t')}\,\mathrm{d}t', \tag{56.5.2}$$

that is

$$G(t) = e^{g(t)}\left[G(0) + \frac{1}{\tau_B}\int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right]. \tag{56.5.3}$$

Step 3: fix $G(0)$ by antiperiodicity. Set $t = T_h = \pi/\omega$ in (56.5.3) and impose $G(T_h) = -G(0)$:

$$-G(0) = e^{g(T_h)}\left[G(0) + \frac{1}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t\right] \tag{56.5.4}$$

$$-G(0) - G(0)e^{g(T_h)} = \frac{e^{g(T_h)}}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t \tag{56.5.5}$$

$$G(0)\left[1 + e^{g(T_h)}\right] = -\frac{e^{g(T_h)}}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t \tag{56.5.6}$$

$$G(0) = -\frac{1}{\tau_B}\,\frac{e^{g(T_h)}}{1 + e^{g(T_h)}}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t. \tag{56.5.7}$$

Step 4: assemble. Substituting (56.5.7) into (56.5.3) and factoring $1/\tau_B$:

$$G(t) = \frac{e^{g(t)}}{\tau_B}\left(-\frac{e^{g(\pi/\omega)}}{1 + e^{g(\pi/\omega)}}\int_0^{\pi/\omega} F(t)e^{-g(t)}\,\mathrm{d}t + \int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right), \tag{SM-42}$$

which is (SM-42). ■

56.6 How the Computation Is Actually Done

The authors state the recipe: integrate (56.5.3) over the half period and substitute the result of the averaging (SM-41) into (SM-38). Explicitly, for prescribed ξ and $\omega\tau_B$:

1. Compute $L(\xi)$ from Eq. (10.2.16) and $u = e^{-\pi/\omega\tau_B}$.
2. Evaluate $F(t)$ from (56.3.2) and $g(t)$ from (56.4.6) on $0 \le t \le \pi/\omega$.
3. Compute the inner integral $\int_0^t Fe^{-g}\mathrm{d}t'$ by quadrature, and hence $G(t)$ from (SM-42).
4. Average: $\overline{G\gamma} = (\omega/\pi)\int_0^{\pi/\omega} G\,\mathrm{d}t$ by (SM-41).
5. Finally $\Delta\eta = \frac{3}{4}\eta\varphi\xi\,\overline{G\gamma}$ by (SM-38).

Three nested quadratures, no differential equations left to integrate. The results are the paper's Figs. 3–5.

56.7 Connection to the Next Chapter

What those figures show, and what they mean physically, is the subject of Chapter 57.

Chapter 57

The Numerical Results: Figures 3, 4 and 5

57.1 Context

Sec. III of the paper closes with three figures, which contain the finite-amplitude answer in graphical form. No new equations appear, but the content is essential and must be understood, since it is what Bacri *et al.* compare their measurements with.

57.2 Figure 3: $\Delta\eta$ versus frequency, at fixed amplitude

The quantity plotted is the relative viscosity

$$\frac{\Delta\eta\big(\xi,\omega\tau_B\big)}{\Delta\eta\big(\infty,0\big)}, \qquad \Delta\eta\big(\infty,0\big)=\tfrac{3}{2}\eta\varphi, \tag{57.2.1}$$

as a function of $\omega\tau_B$, for $\xi = 2, 5, 10, 15$. The normalisation is the saturation viscosity in a permanent field, Eq. (SM-4).

Three features, all of which we can now explain:

1. **Every curve starts positive and ends negative.** The zero-frequency value is the stationary result (SM-2); the high-frequency tail is negative, as Sec. 54.6(b) showed.

2. **The crossing frequency ω_0 decreases as ξ grows.** Stronger fields make the decrease of $\Delta\eta(\omega\tau_B)$ steeper. The reason is the feedback term of Sec. 51.3: at large ξ the effective relaxation rate of the transverse magnetisation is $(1+\frac{1}{2}\xi F)/\tau_B$, substantially faster than $1/\tau_B$. A faster effective relaxation means the phase lag reaches the critical value at a lower frequency.

3. **The minimum deepens with ξ.** At $\xi = 15$ the negative excursion approaches $-\Delta\eta(\infty,0)$, i.e. $\Delta\eta \approx -\frac{3}{2}\eta\varphi$ — comparable in magnitude to the entire saturation increase produced by a static field.

57.3 Figure 4: $\Delta\eta$ versus amplitude, at fixed frequency

Here $\Delta\eta/\Delta\eta(\infty,0)$ is plotted against ξ for $\omega\tau_B = 0, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2$. The striking feature, which the authors highlight, is that the dependence on ξ is *non-monotonic*: for $0 <$

$\omega\tau_B < 1$ every curve rises to a maximum and then falls into the negative region as the field is increased further.

Why non-monotonic? Two competing effects, both growing with ξ:

- the magnitude of the magnetisation, hence of the torque, grows — this pushes $|\Delta\eta|$ up;
- the effective relaxation rate $(1 + \frac{1}{2}\xi F)/\tau_B$ grows, which reduces the phase lag and eventually carries the system past the critical phase — this drives $\Delta\eta$ negative.

At small ξ the first wins; at large ξ the second does. The crossover is the maximum. Only the topmost curve, $\omega\tau_B = 0$, is monotonic — and that one is just the stationary formula (SM-2).

A prediction worth noting. In sufficiently strong fields the rotational viscosity is negative at *any* nonzero frequency. Only in the strictly stationary field is $\Delta\eta$ positive for all ξ, as thermodynamics requires (Sec. 15.6).

57.4 Figure 5: the map of viscosity

The most useful representation is the plane $(\xi, \omega\tau_B)$ with isolines $\Delta\eta = \text{const}$. The solid line $\Delta\eta = 0$ divides the plane into a positive region on the left (weak fields, low frequencies) and a negative region on the right. The numbers on the isolines are values of $\Delta\eta/\Delta\eta(\infty, 0)$, so that for instance the isoline labelled -1 corresponds to $\Delta\eta = -\frac{3}{2}\eta\varphi$.

This diagram is the direct theoretical counterpart of Fig. 4 of Bacri *et al.*, which plots the same isolines computed from the effective-field theory, and of their Fig. 5, which plots the measured ones. The comparison is the subject of Volume VII.

Chapter 58

Section IV of the Paper: Can the Total Viscosity Be Negative?

58.1 Context

The concluding section of Shliomis & Morozov asks the natural question: since $\Delta\eta < 0$ is possible, can it be large enough to make the *total* viscosity negative,

$$\eta_{\text{total}} = \eta + \Delta\eta < 0\ ? \tag{58.1.1}$$

The answer determines whether the effect is a curiosity or a route to a self-pumping fluid.

58.2 The condition, in the notation of Figure 5

Step 1. The condition $\eta_{\text{total}} = 0$ means $\Delta\eta = -\eta$.

Step 2. Divide by the normalisation used in Fig. 5, $\Delta\eta(\infty, 0) = \frac{3}{2}\eta\varphi$:

$$\frac{\Delta\eta\big(\xi, \omega\tau_B\big)}{\Delta\eta\big(\infty, 0\big)} = \frac{-\eta}{\frac{3}{2}\eta\varphi} = -\frac{2}{3\varphi}. \tag{58.2.1}$$

This is the unnumbered relation printed in the paper's Sec. IV.

Step 3: the number. At the realistic concentration $\varphi = 0.2$,

$$-\frac{2}{3 \times 0.2} = -\frac{2}{0.6} = -\frac{10}{3} \approx -3.33, \tag{58.2.2}$$

so $\eta_{\text{total}} = 0$ corresponds to the isoline $-10/3$ of Fig. 5. Since the figure's most negative plotted isoline is -1, the required point lies far outside the plotted region — in the domain of very high ξ and $\omega\tau_B$.

58.3 The Stability Argument

The authors approach the same question from a second direction, which is worth recording because it converts a statement about viscosity into a statement about *instability*.

The idea. A fluid with $\eta_{\text{total}} < 0$ cannot remain at rest: any infinitesimal disturbance grows rather than decays, because the "viscous" term now feeds energy into the disturbance instead of dissipating it. Therefore the neutral curve of the stability of mechanical equilibrium in the $(\xi, \omega\tau_B)$ plane is exactly the isoline $\eta_{\text{total}} = 0$.

The set-up. Reference 5 of the paper studies a ferrofluid in a plane channel between rigid walls at $x = 0$ and $x = l$, with $H_x = H_0 \cos \omega t$. For the one-dimensional perturbation

$$\boldsymbol{v} = \big[0,\ v(x,t),\ 0\big] \qquad \Longrightarrow \qquad \boldsymbol{\Omega} = \tfrac{1}{2}\nabla \times \boldsymbol{v} = \left(0,\ 0,\ \tfrac{1}{2}\frac{\partial v}{\partial x}\right), \tag{58.3.1}$$

where the vorticity was computed from the definition (3.4.1): for $\boldsymbol{v} = v(x,t)\hat{\boldsymbol{y}}$,

$$\nabla \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z},\ \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x},\ \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right) = \left(0, 0, \frac{\partial v}{\partial x}\right). \checkmark \tag{58.3.2}$$

The mode structure. Seeking solutions of the form

$$v = w(t) \sin qx, \qquad M_y = f(t) \cos qx, \tag{58.3.3}$$

the no-slip boundary conditions $v(0,t) = v(l,t) = 0$ require $\sin ql = 0$, hence (with l taken as the unit of length)

$$q = \pi r, \qquad r = 1, 2, 3, \ldots \tag{58.3.4}$$

Derivation of the spectrum: $\sin(0) = 0$ automatically; $\sin(ql) = 0$ requires ql to be an integer multiple of π.

Which mode goes unstable first. The viscous damping rate of a mode is proportional to q^2 (from $\eta\nabla^2\boldsymbol{v}$, which for $\sin qx$ gives $-\eta q^2$), so the mode with the smallest q is the least damped and the first to become unstable. That is $r = 1$, i.e. $q = \pi$.

Why this mode matters physically. With $v = w(t) \sin \pi x$ on $0 < x < 1$, the velocity has the same sign across the whole channel, so the volume flux

$$\int_0^1 v \,\mathrm{d}x = w(t) \int_0^1 \sin \pi x \,\mathrm{d}x = w(t) \left[-\frac{\cos \pi x}{\pi}\right]_0^1 = w(t)\frac{2}{\pi} \tag{58.3.5}$$

is nonzero. The instability therefore produces a *net flow through the channel*: as the authors put it, if ξ and $\omega\tau_B$ lie in the unstable region the entire system works as a pump, driven by the alternating field with no moving parts and no applied pressure gradient.

58.4 The Critical Values, and Why They Are Out of Reach

For $q = \pi$ and $\varphi = 0.2$ the minimum on the neutral curve is found at

$$\xi_c = 126, \qquad \omega_c\tau_B \text{ correspondingly large.} \tag{58.4.1}$$

Bacri *et al.* quote the companion value $\omega_c\tau_B = 20$.

What $\xi = 126$ means physically. For the experimental grains, $\xi = 5.25 \times 10^{-3} H(\mathrm{Oe})$, so

$$H_c = \frac{126}{5.25 \times 10^{-3}} = 2.4 \times 10^4 \text{ Oe} = 24 \text{ kOe}. \tag{58.4.2}$$

That is a large but not impossible laboratory field. The real obstacle is different, and both papers identify it: such a value of ξ could be reached at a modest field only by using *larger* particles (since $\xi \propto d^3$), and larger particles bring two fatal problems — they aggregate, destroying the colloidal stability, and they make the fluid non-Newtonian. Moreover, at 10^4 Oe the assumption that the moment is rigidly frozen in the crystal begins to fail: the field can rotate the moment inside the grain, which decouples the magnetic from the mechanical degrees of freedom and switches the entire effect off. The authors state exactly this: the required ξ_c demands an amplitude at which the particles can no longer be treated as rigid dipoles.

The honest conclusion. Negative *rotational* viscosity is real, measurable, and was measured. Negative *total* viscosity — a spontaneously pumping ferrofluid — remains, on the basis of this theory, out of reach for practical ferrocolloids. What is within reach is the tunability that Bacri *et al.* emphasise: a viscosity adjustable between 220 and 50 cP by changing only the frequency of a fixed-amplitude field, which they suggest could form the basis of adaptive dampers.

Chapter 59

Status and Road Map

59.1 What has been established in Volume VI

Equation	Statement	Where derived
(SM-35)	nonlinear magnetisation equation at arbitrary amplitude; $\xi(t) = \xi\gamma(t)$	Ch. 50
(50.4.1)	$M_x = nmF$, $M_y = nm\Omega\tau_B G$	Sec. 50.4
(SM-36)	$\tau_B\dot{F} = L(\xi\gamma) - F$; $\tau_B\dot{G} = F - (1 + \frac{1}{2}\xi\gamma F)G$	Ch. 51
(SM-37)	$\overline{(\boldsymbol{M}\times\boldsymbol{H})_z} = -3\eta\varphi\Omega\xi\overline{G\gamma}$	Ch. 52
(SM-38)	$\Delta\eta = \frac{3}{4}\eta\varphi\xi\overline{G\gamma}$	Ch. 53
(SM-39)	definition of the period average	Sec. 53.1
(53.4.2)	$(\Omega - \Theta)/\Omega = \frac{\xi}{2}\overline{G\gamma}$, recovering (SM-34) at any amplitude	Sec. 53.4
(SM-40)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\big(2 - \tanh^2\varepsilon - 2\tanh\varepsilon/\varepsilon\big)$; misprint corrected and verified against $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, -0.105	Ch. 54
(SM-41)	half-period reduction, from antiperiodicity	Ch. 55
(SM-42)	formal solution for G by integrating factor	Sec. 56.5
(SM-43)	$F(t)$ and $g(t)$ in closed form	Secs. 56.3–56.4
Figs. 3–5	frequency, amplitude and map representations; non-monotonic $\Delta\eta(\xi)$ explained	Ch. 57
Sec. IV	$\eta_{\text{total}} = 0 \Leftrightarrow$ isoline $-2/3\varphi = -10/3$; $q = \pi$; the pump; $\xi_c = 126$	Ch. 58

59.2 Cumulative progress

Shliomis & Morozov (1994): complete. All 43 numbered equations, together with the unnumbered relations of Secs. II–IV, have now been derived from first principles.

Bacri, Perzynski, Shliomis & Burde (1995): Eqs. (B-1), (B-2), (B-3), (B-4) and (B-10) complete.

59.3 What remains: Volume VII

The final volume treats the experimental paper's own theoretical contribution — the effective-field method, which Bacri *et al.* adopt precisely because the phenomenological equation (SM-7) fails for finite ξ *and* finite $\omega\tau_B$ simultaneously (the limitation identified in Sec. 13.4):

- **(B-5)** $\boldsymbol{M} = \varphi M_s L(\xi_e)\boldsymbol{\xi}_e/\xi_e$: the effective-field ansatz, derived from the Fokker–Planck equation (11.6.1);
- **(B-6)** the evolution equation for $\boldsymbol{\xi}_e$, including the L_3 term;
- **(B-7)** the zero-approximation equation for ξ_e in the small parameter $\Omega\tau_B$;
- **(B-8)** the equation for the function F;
- **(B-9)** the mean magnetic torque and the function $g = \frac{1}{2}\xi_0\overline{\cos\omega t\, L(\xi_e)F(\xi_e)}$;
- the experimental analysis: the reduced viscosity $\eta_r(H, f)$, the calibrations $\xi_0 = 5.25 \times 10^{-3}H(\mathrm{Oe})$ and $\omega\tau_B = 10^{-2}f(\mathrm{Hz})$, the isoline comparison between Figs. 4 and 5, the observed 25% viscosity reduction, and the critical values $\xi_c = 126$, $\omega_c\tau_B = 20$.

End of Volume VI. Volume VII — the final volume — begins with Eq. (B-5).